

Exemplar_Course 4 TikTok project lab

July 17, 2026

1 TikTok Project

Course 3 - The Power of Statistics

You are a data professional at TikTok. The current project is reaching its midpoint; a project proposal, Python coding work, and exploratory data analysis have all been completed.

The team has reviewed the results of the exploratory data analysis and the previous executive summary the team prepared. You received an email from Orion Rainier, Data Scientist at TikTok, with your next assignment: determine and conduct the necessary hypothesis tests and statistical analysis for the TikTok classification project.

A notebook was structured and prepared to help you in this project. Please complete the following questions.

2 Course 3 End-of-course project: Data exploration and hypothesis testing

In this activity, you will explore the data provided and conduct a hypothesis testing.

The purpose of this project is to demonstrate knowledge of how to prepare, create, and analyze hypothesis tests.

The goal is to apply descriptive and inferential statistics, probability distributions, and hypothesis testing in Python.

This activity has three parts:

Part 1: Imports and data loading * What data packages will be necessary for hypothesis testing?

Part 2: Conduct hypothesis testing * How will descriptive statistics help you analyze your data?

- How will you formulate your null hypothesis and alternative hypothesis?

Part 3: Communicate insights with stakeholders

- What key business insight(s) emerge from your hypothesis test?
- What business recommendations do you propose based on your results?

Follow the instructions and answer the questions below to complete the activity. Then, complete an executive summary using the questions listed on the PACE Strategy Document.

Be sure to complete this activity before moving on. The next course item will provide you with a completed exemplar to compare to your own work.

3 Data exploration and hypothesis testing

4 PACE stages

Throughout these project notebooks, you'll see references to the problem-solving framework PACE. The following notebook components are labeled with the respective PACE stage: Plan, Analyze, Construct, and Execute.

4.1 PACE: Plan

Consider the questions in your PACE Strategy Document and those below to craft your response.

1. What is your research question for this data project? Later on, you will need to formulate the null and alternative hypotheses as the first step of your hypothesis test. Consider your research question now, at the start of this task.

Exemplar response:

There are a few possible ways to frame the research question. For example:

- 1) Do videos from verified accounts and videos unverified accounts have different average view counts?
- 2) Is there a relationship between the account being verified and the associated videos' view counts?

Complete the following steps to perform statistical analysis of your data:

4.1.1 Task 1. Imports and Data Loading

Import packages and libraries needed to compute descriptive statistics and conduct a hypothesis test.

Hint:

Be sure to import `pandas`, `numpy`, `matplotlib.pyplot`, `seaborn`, and `scipy`.

```
[1]: # Import packages for data manipulation
import pandas as pd
import numpy as np

# Import packages for data visualization
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Import packages for statistical analysis/hypothesis testing
from scipy import stats
```

Load the dataset.

Note: As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the .csv file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

```
[2]: # Load dataset into dataframe
data = pd.read_csv("tiktok_dataset.csv")
```

4.2 PACE: Analyze and Construct

Consider the questions in your PACE Strategy Document and those below to craft your response:

1. Data professionals use descriptive statistics for Exploratory Data Analysis. How can computing descriptive statistics help you learn more about your data in this stage of your analysis?

Exemplar response:

In general, descriptive statistics are useful because they let you quickly explore and understand large amounts of data. In this case, computing descriptive statistics helps you quickly compute the mean values of video_view_count for each group of verified_status in the sample data.

4.2.1 Task 2. Data exploration

Use descriptive statistics to conduct Exploratory Data Analysis (EDA).

Hint:

Refer back to *Self Review Descriptive Statistics* for this step-by-step process.

Inspect the first five rows of the dataframe.

```
[3]: # Display first few rows
data.head()
```

```
[3]:  # claim_status    video_id  video_duration_sec  \
0  1      claim    7017666017          59
1  2      claim    4014381136          32
2  3      claim    9859838091          31
3  4      claim    1866847991          25
4  5      claim    7105231098          19

      video_transcription_text  verified_status  \
0  someone shared with me that drone deliveries a...  not verified
1  someone shared with me that there are more mic...  not verified
2  someone shared with me that american industria...  not verified
3  someone shared with me that the metro of st. p...  not verified
```

4 someone shared with me that the number of busi... not verified

	author_ban_status	video_view_count	video_like_count	video_share_count	\
0	under review	343296.0	19425.0	241.0	
1	active	140877.0	77355.0	19034.0	
2	active	902185.0	97690.0	2858.0	
3	active	437506.0	239954.0	34812.0	
4	active	56167.0	34987.0	4110.0	

	video_download_count	video_comment_count
0	1.0	0.0
1	1161.0	684.0
2	833.0	329.0
3	1234.0	584.0
4	547.0	152.0

```
[4]: # Generate a table of descriptive statistics about the data
data.describe()
```

```
[4]:
```

	#	video_id	video_duration_sec	video_view_count	\
count	19382.000000	1.938200e+04	19382.000000	19084.000000	
mean	9691.500000	5.627454e+09	32.421732	254708.558688	
std	5595.245794	2.536440e+09	16.229967	322893.280814	
min	1.000000	1.234959e+09	5.000000	20.000000	
25%	4846.250000	3.430417e+09	18.000000	4942.500000	
50%	9691.500000	5.618664e+09	32.000000	9954.500000	
75%	14536.750000	7.843960e+09	47.000000	504327.000000	
max	19382.000000	9.999873e+09	60.000000	999817.000000	

	video_like_count	video_share_count	video_download_count	\
count	19084.000000	19084.000000	19084.000000	
mean	84304.636030	16735.248323	1049.429627	
std	133420.546814	32036.174350	2004.299894	
min	0.000000	0.000000	0.000000	
25%	810.750000	115.000000	7.000000	
50%	3403.500000	717.000000	46.000000	
75%	125020.000000	18222.000000	1156.250000	
max	657830.000000	256130.000000	14994.000000	

	video_comment_count
count	19084.000000
mean	349.312146
std	799.638865
min	0.000000
25%	1.000000
50%	9.000000
75%	292.000000

```
max          9599.000000
```

Check for and handle missing values.

```
[5]: # Check for missing values
data.isna().sum()
```

```
[5]: #
claim_status      298
video_id          0
video_duration_sec 0
video_transcription_text 298
verified_status   0
author_ban_status 0
video_view_count  298
video_like_count  298
video_share_count 298
video_download_count 298
video_comment_count 298
dtype: int64
```

```
[6]: # Drop rows with missing values
data = data.dropna(axis=0)
```

```
[7]: # Display first few rows after handling missing values
data.head()
```

```
[7]: # claim_status    video_id  video_duration_sec  \
0  1      claim    7017666017          59
1  2      claim    4014381136          32
2  3      claim    9859838091          31
3  4      claim    1866847991          25
4  5      claim    7105231098          19

      video_transcription_text  verified_status  \
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      author_ban_status  video_view_count  video_like_count  video_share_count  \
0      under review      343296.0      19425.0      241.0
1      active          140877.0      77355.0      19034.0
2      active          902185.0      97690.0      2858.0
3      active          437506.0     239954.0     34812.0
4      active          56167.0      34987.0      4110.0
```

	video_download_count	video_comment_count
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You are interested in the relationship between `verified_status` and `video_view_count`. One approach is to examine the mean values of `video_view_count` for each group of `verified_status` in the sample data.

```
[8]: # Compute the mean `video_view_count` for each group in `verified_status`
    ## YOUR CODE HERE ##
    data.groupby("verified_status")["video_view_count"].mean()
```

```
[8]: verified_status
not verified    265663.785339
verified       91439.164167
Name: video_view_count, dtype: float64
```

4.2.2 Task 3. Hypothesis testing

Before you conduct your hypothesis test, consider the following questions where applicable to complete your code response:

1. Recall the difference between the null hypothesis and the alternative hypotheses. What are your hypotheses for this data project?

Exemplar response:

- **Null hypothesis:** There is no difference in number of views between TikTok videos posted by verified accounts and TikTok videos posted by unverified accounts (any observed difference in the sample data is due to chance or sampling variability).
- **Alternative hypothesis:** There is a difference in number of views between TikTok videos posted by verified accounts and TikTok videos posted by unverified accounts (any observed difference in the sample data is due to an actual difference in the corresponding population means).

Your goal in this step is to conduct a two-sample t-test. Recall the steps for conducting a hypothesis test:

1. State the null hypothesis and the alternative hypothesis
2. Choose a significance level
3. Find the p-value
4. Reject or fail to reject the null hypothesis

H_0 : There is no difference in number of views between TikTok videos posted by verified accounts and TikTok videos posted by unverified accounts (any observed difference in the sample data is due to chance or sampling variability).

H_A : There is a difference in number of views between TikTok videos posted by verified accounts and TikTok videos posted by unverified accounts (any observed difference in the sample data is due to an actual difference in the corresponding population means).

You choose 5% as the significance level and proceed with a two-sample t-test.

```
[9]: # Conduct a two-sample t-test to compare means
    ### YOUR CODE HERE ###

    # Save each sample in a variable
    not_verified = data[data["verified_status"] == "not_
    ↪verified"]["video_view_count"]
    verified = data[data["verified_status"] == "verified"]["video_view_count"]

    # Implement a t-test using the two samples
    stats.ttest_ind(a=not_verified, b=verified, equal_var=False)
```

```
[9]: Ttest_indResult(statistic=25.499441780633777, pvalue=2.6088823687177823e-120)
```

Exemplar response:

Since the p-value is extremely small (much smaller than the significance level of 5%), you reject the null hypothesis. You conclude that there **is** a statistically significant difference in the mean video view count between verified and unverified accounts on TikTok.

5 PACE: Execute

Consider the questions in your PACE Strategy Document to reflect on the Execute stage.

5.0.1 Task 4. Communicate insights with stakeholders

Ask yourself the following question:

- What business insight(s) can you draw from the result of your hypothesis test?

Exemplar response:

The analysis shows that there is a statistically significant difference in the average view counts between videos from verified accounts and videos from unverified accounts. This suggests there might be fundamental behavioral differences between these two groups of accounts.

It would be interesting to investigate the root cause of this behavioral difference. For example, do unverified accounts tend to post more clickbait-y videos? Or are unverified accounts associated with spam bots that help inflate view counts?

The next step will be to build a regression model on `verified_status`. A regression model is the natural next step because the end goal is to make predictions on claim status. A regression model for `verified_status` can help analyze user behavior in this group of verified users. Technical note

to prepare regression model: because the data is skewed, and there is a significant difference in account types, it will be key to build a logistic regression model.

Congratulations! You've completed this lab. However, you may not notice a green check mark next to this item on Coursera's platform. Please continue your progress regardless of the check mark. Just click on the "save" icon at the top of this notebook to ensure your work has been logged.